Entrepreneurial Finance | Term paper

Digital credit access and financing constraints amongst firms: Evidence from India and peer economies

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ABSTRACT

Small firms are often denied credit because lenders cannot observe their quality. Digital finance is widely argued to ease this constraint, but evidence on India is mostly descriptive. This study asks whether digital access goes with fewer credit-constrained firms and where India stands relative to peer economies. We build a country-level panel of 215 World Bank Enterprise Survey observations (142 economies, 2013–2024), define credit constraint as the share of firms that are fully or partially credit constrained, and test whether digital-access indicators help explain it beyond traditional financial depth. A ridge baseline on income, private credit, lending spread and macro controls reaches a country-held-out R^2 of 0.505 (MAE 10.2 percentage points). Adding digital features and using gradient boosting raises R^2 to 0.522 and lowers MAE to 10.0 percentage points, a modest gain; income per capita remains the dominant predictor. India (2022) has 34.5% of firms constrained (10.5% fully), against a sample median of 26.7%. The digital model predicts 31.8% for India out of sample, and 27.3% if India's digital indicators reached the sample top quartile. Results are associational, not causal. The findings support bank–FinTech data sharing and digitisation of firm records, and point to firm-level and quasi-experimental work as the next step.

1. Introduction

Access to external finance is a binding constraint on firm growth, especially for micro, small and medium enterprises (MSMEs) that lack hard collateral, audited accounts and long credit histories (Lu et al., 2022; Asian Institute of Management Rizalino Navarro Policy Center for Competitiveness [AIM RNP], 2019). MSMEs are most of the world's firms, yet they are systematically underserved. The Global Partnership for Financial Inclusion and the International Finance Corporation (GPFI & IFC, 2024) estimate the global MSME credit gap at close to US$5 trillion, and industry estimates put the gap for India near US$500 billion (ETV Bharat, 2024).

Three frictions explain why good firms still go unfunded. First, borrowers are *opaque*: young, asset-light firms have no audited accounts, so a lender cannot tell good from bad. Second, they have *no hard collateral*: banks price risk on land and fixed assets, so intangible-heavy SMEs are rationed. Third, they have a *thin credit history*: first-time borrowers have no bureau record, and the screening cost per small loan is high. Digital credit, meaning lending screened with transaction, mobile and platform data, offers a way round all three. Lenders can use payments, platform and tax data instead of collateral, screen cheaply and lend at scale. A growing literature links digital finance to eased financing constraints and, downstream, to more breakthrough innovation, green innovation and resilience (Feng et al., 2023; Han & Zhang, 2023; Yao & Yang, 2022).

Most of that evidence, however, relies on constructed indices of constraint, such as the Kaplan–Zingales and Whited–Wu indices, and on Chinese listed firms. Evidence on India is mostly

descriptive. A literature review alone cannot test these claims. This paper therefore moves from review to measurement and focuses on India. The central question is stated below.

Central research question. *Does digital access go with fewer credit-constrained firms, and where does India stand?*

RQ1. How constrained are Indian firms relative to peer economies?

RQ2. Do digital-access indicators explain credit constraint beyond traditional financial depth?

RQ3. What does a simple machine-learning model imply for India's position and its digital headroom?

We answer these questions with open data. We assemble a country-level panel from the World Bank, measure credit constraint from actual loan outcomes, and compare a traditional-variable baseline with models that add digital-access indicators. The contribution is predictive benchmarking, not causal identification. The rest of the paper is organised as follows. Section 1.1 traces the history of digital credit and Section 1.2 defines the terms used. Section 2 reviews the theory and literature. Section 3 describes the data and methodology. Section 4 reports the findings. Section 5 discusses academic, managerial and policy implications, limitations and future scope. Section 6 concludes.

1.1. A short history of digital credit

Digital credit did not appear at once. It built up over four decades, and each wave removed one blind spot that kept lenders from seeing small borrowers (Table 1). The starting point is theoretical: Stiglitz and Weiss (1981) showed that information problems can make lenders ration credit even in equilibrium. Microfinance, formalised by Grameen Bank in 1983, showed that poor borrowers can be reached by substituting group liability and relationships for collateral (Morduch, 1999). Peer-to-peer lending arrived in 2005 with Zopa in the United Kingdom, and mobile money followed in 2007 with M-Pesa in Kenya, which created a digital record of payments for people outside the banking system (Jack & Suri, 2014).

India's public digital stack then compressed the same sequence. Aadhaar (2009) provided digital identity; Jan Dhan accounts (2014) brought mass inclusion, and account ownership among Indian adults rose from 53% in 2014 to 77.5% in 2021 (Demirgüç-Kunt et al., 2022); the Unified Payments Interface (UPI, 2016) made instant payments routine; and the Account Aggregator framework (2021) created a consent mechanism for sharing financial data. The pattern is consistent: each wave removed one lender blind spot, namely identity, payments history, or consent to share data.

Table 1
Milestones in the history of digital credit

Year	Milestone	Where	Lender blind spot addressed
1981	Credit rationing theory (Stiglitz & Weiss)	Theory	Explains why information gaps cause rationing
1983	Microfinance scales (Grameen Bank)	Bangladesh	No collateral: group lending replaces it
2005	Peer-to-peer lending (Zopa)	United Kingdom	Intermediary cost: platforms match lenders and borrowers
2007	Mobile money (M-Pesa)	Kenya	No payment record: digital transactions create one
2009	Digital identity (Aadhaar)	India	Identity: unique, verifiable borrower ID
2014	Mass inclusion (Jan Dhan accounts)	India	No account: brings households into the banking system
2016	Instant payments (UPI)	India	Payments history: rich, real-time transaction trail

2021	Data-sharing consent (Account Aggregator)	India	Consent: borrower-approved data portability

1.2. Definitions of key terms

The paper and the accompanying slides use a number of technical terms from entrepreneurial finance, development finance and machine learning. Table 2 defines them as they are used here.

Table 2
Definitions of terms used in the report and slides

Term	Definition as used in this paper
Finance and firm terms	
MSME / SME	Micro, small and medium enterprise. Small firms are the focus because they are the group most affected by information problems.
MSME credit gap	The difference between the credit MSMEs need and the credit they receive from formal lenders (GPFI & IFC, 2024).
Collateral	An asset pledged to secure a loan, typically land or fixed assets. Intangible-heavy firms often lack it.
Credit history / credit bureau	A record of past borrowing and repayment held by a credit bureau. A thin credit history means little or no record exists.
Opaque borrower	A borrower whose quality a lender cannot observe, for example because it lacks audited accounts.
Screening cost	The cost to a lender of assessing a borrower's creditworthiness. It is high relative to loan size for small loans.
Digital credit	Lending in which approval is screened with digital data such as transactions, mobile, e-commerce, platform and tax records.
Alternative data	Non-traditional information used to assess borrowers in place of collateral and bureau records.
FinTech	A technology-based financial firm, here one that lends or screens using digital data.
Bank–FinTech coopetition	A relationship in which banks and FinTechs both cooperate and compete: banks bring capital, FinTechs bring data and speed (Larsson et al., 2024).
Microfinance	Small-scale financial services for borrowers excluded from formal banking, often delivered through group lending.
Peer-to-peer (P2P) lending	Online platforms that match individual or institutional lenders directly with borrowers.
Mobile money	Financial services, mainly payments and transfers, delivered through mobile phones, for example M-Pesa.
Digital public infrastructure (DPI)	Publicly built digital systems for identity, payments and data sharing. India's stack comprises Aadhaar, Jan Dhan, UPI, GST data and the Account Aggregator.
Aadhaar	India's biometric digital identity system.
Jan Dhan	India's national programme (2014) to open basic bank accounts for unbanked households.
UPI	Unified Payments Interface, India's instant, interoperable mobile payments system (2016).
Account Aggregator (AA)	India's consent-based framework (live 2021) that lets individuals and firms share their financial data with lenders.
GST data	Records from India's Goods and Services Tax filings, which show firms' sales and can be used in screening.
Theory and measurement terms	
Information asymmetry	A situation in which one party (the borrower) knows more than the other (the lender).
Adverse selection / moral hazard	Adverse selection: higher interest rates attract riskier borrowers. Moral hazard: higher rates push borrowers toward riskier projects (Stiglitz & Weiss, 1981).
Credit rationing	Lenders refuse or cut loans instead of raising interest rates; it can persist in equilibrium (Stiglitz & Weiss, 1981).
Pecking order	The theory that external funds cost more than internal funds, so firms prefer internal finance first (Myers, 1984).
Financing constraint	A wedge between the internal and external cost of funds that makes firms forgo profitable investment.

Forgone investment	Investment a firm would have made had finance been available: lower innovation, growth and resilience.
KZ and WW indices	Index-based proxies for financing constraints built from firm accounting data (Kaplan & Zingales, 1997; Whited & Wu, 2006). Both are imperfect proxies.
Cash-flow sensitivity	The responsiveness of investment to internal cash flow, used (and debated) as a constraint measure (Fazzari et al., 1988; Kaplan & Zingales, 2000).
Fully / partially constrained	Following Kuntchev et al. (2013), firms are classed from loan outcomes as not needing credit, unconstrained, partially constrained, or fully constrained. We use the percentage of partially plus fully constrained firms.
Enterprise Surveys (WBES)	The World Bank's firm-level surveys, whose published country aggregates supply our outcome.
WDI and Global Findex	World Development Indicators (macro and digital indicators) and the Global Findex (survey of account ownership, triennial).
Private credit / GDP	Domestic credit to the private sector as a share of GDP; a measure of financial depth.
Lending spread	Lending rate minus deposit rate; a measure of the cost of intermediation.
Secure internet servers	Servers using encryption, per million people; a proxy for digital-commerce infrastructure.
Account ownership	Share of adults (15+) with an account at a financial institution or a mobile money provider.
Modelling terms	
Ridge regression	A linear regression with an L2 penalty on coefficients that limits overfitting and handles correlated predictors (Hoerl & Kennard, 1970).
Random forest	An ensemble of decision trees trained on resampled data whose predictions are averaged (Breiman, 2001).
Gradient boosting	An ensemble that builds shallow trees sequentially, each correcting the errors of the previous ones (Friedman, 2001).
Country-held-out (group) cross-validation	Cross-validation in which all observations of a country fall in the same fold, so every score is out-of-country and free of leakage.
R^2 and MAE	R^2: share of variance in the outcome explained out of sample. MAE: mean absolute error, here in percentage points (pp).
Permutation importance	The fall in model accuracy when one predictor's values are randomly shuffled; a larger fall means a more important predictor.
Counterfactual	A what-if prediction: here, India's predicted constraint if its digital indicators reached the sample top quartile.
Collinearity	High correlation among predictors, which makes it hard to separate their individual effects (digital reach and income are collinear here).
Associational vs causal; ecological	An associational result shows co-movement without proving cause. An ecological conclusion is drawn from group (country) aggregates and may not hold for individual firms.

2. Theoretical background and literature review

2.1. From information asymmetry to financing constraints

The theoretical starting point is information asymmetry: the borrower knows more about the project and its own quality than the lender. Stiglitz and Weiss (1981) showed that adverse selection and moral hazard make lenders ration credit even in equilibrium, because a higher interest rate worsens the mix of borrowers rather than clearing the market. A related strand, the pecking-order view, holds that external funds cost more than internal funds because outsiders discount what they cannot observe (Myers, 1984). The consequence is a financing constraint, a wedge between the internal and external cost of funds, and with it forgone investment: lower innovation, growth and resilience.

Measuring that constraint is contested. One tradition infers constraints from the sensitivity of investment to cash flow (Fazzari et al., 1988), but Kaplan and Zingales (1997, 2000) show that such sensitivities are not valid measures. Index approaches, such as the Kaplan–Zingales (KZ) and Whited–Wu (WW) indices, offer convenient proxies, though Whited and Wu (2006) themselves

emphasise their limits. A third, outcome-based tradition classifies firms from what actually happened when they sought credit. Kuntchev et al. (2013) argue that perception-based answers about "access to finance" are unreliable and instead sort firms into those not needing credit, unconstrained, partially constrained and fully constrained. The World Bank Enterprise Surveys publish aggregate shares that follow this logic, and we adopt it because it avoids the weaknesses of index proxies.

2.2. How digital finance eases constraints

The literature identifies three mechanisms. The first is *alternative data*. Transaction, mobile, e-commerce and tax records can substitute for collateral and bureau history. Studies on Chinese data find that digital inclusive finance lowers SME constraints, with local banks and digital inclusion working together (Lu et al., 2022) and with particular benefit to technology-based small and medium enterprises (Wang & Chen, 2022). The second is *transparency*. Digitised firm records and platform activity make firms visible, and firm characteristics such as record-keeping and formality are strongly tied to credit constraints (AIM RNP, 2019). The third is *bank–FinTech coopetition*: banks bring capital and FinTechs bring data and speed, and the two co-exist in a market ecosystem that is both cooperative and competitive (Larsson et al., 2024).

Eased constraints are in turn channelled into other outcomes. Yao and Yang (2022) find that digital finance boosts SME innovation by easing financing constraints among Chinese GEM-listed companies. Han and Zhang (2023) link digital finance to enterprise green innovation, and Feng et al. (2023) show that digital finance changes the asset allocation of SMEs, with financing constraints as the mediator, which we describe as asset discipline. For India, the public digital stack of Aadhaar, Jan Dhan, UPI, GST data and the Account Aggregator arguably does part of what collateral does elsewhere. Yet most of the causal evidence so far comes from China, not India.

2.3. Research gap

Three gaps follow. The literature relies on KZ and WW index proxies rather than outcome-based measures; Chinese listed firms dominate the evidence; and there is little benchmarking of India against peer economies. Our work addresses these gaps by (i) using an outcome-based measure of constraint, (ii) focusing on India, and (iii) adding a simple machine-learning comparison to test whether digital indicators carry explanatory power beyond traditional financial depth. The three research questions in Section 1 operationalise this agenda.

3. Methodology

This section describes the data, the models, and the validation and interpretation procedures. All code is in a single notebook (*analysis.ipynb*), and every number in the report and slides is generated from it.

3.1. Data

Source and outcome. All data are open and obtained through the World Bank's public World Development Indicators (WDI) API. The outcome is the percentage of firms that are fully plus partially credit constrained, taken from the Enterprise Surveys. The analysis sample has 215 country-year observations from 142 economies, with survey years from 2013 to 2024. Mean shares are 13.8% fully and 16.0% partially constrained, so 29.8% (about 30%) of firms are constrained on average (Table 4).

Predictors. Predictors fall into two groups (Table 3). *Traditional* variables capture income, financial depth and the macro environment. *Digital* variables capture population-level digital reach. Account

ownership comes from the Global Findex, which is triennial; the latest wave is carried forward within each country. Income per capita and secure servers are log transformed. Missing predictor values are median-imputed inside each cross-validation fold.

Table 3
Variables used in the analysis

Variable	Group	Definition	Source
% firms constrained	Outcome	Fully + partially credit constrained firms (%)	Enterprise Surveys
Log GDP per capita	Traditional	Natural log of GDP per capita	WDI
Private credit / GDP	Traditional	Domestic credit to the private sector (% of GDP)	WDI
Lending spread	Traditional	Lending rate minus deposit rate (pp)	WDI
Inflation	Traditional	Annual consumer price inflation (%)	WDI
GDP growth	Traditional	Annual real GDP growth (%)	WDI
Internet users	Digital	Individuals using the internet (% of population)	WDI
Mobile subscriptions	Digital	Mobile cellular subscriptions (per 100 people)	WDI
Broadband subscriptions	Digital	Fixed broadband subscriptions (per 100 people)	WDI
Log secure servers	Digital	Log(1 + secure internet servers per million people)	WDI
Account ownership	Digital	Adults with an account at a financial institution or mobile money provider (%)	Global Findex

Table 4
Descriptive statistics of the analysis sample

Variable	N	Mean	SD	Min	Max
% firms constrained	215	29.80	18.37	0.52	79.37
% fully constrained	215	13.76	12.10	0.00	57.16
% partially constrained	215	16.04	10.88	0.39	66.67
Log GDP per capita	214	8.64	1.31	5.57	11.56
Private credit / GDP	196	51.35	38.66	3.24	248.79
Lending spread	116	6.42	5.12	−1.92	35.46
Inflation	210	5.40	9.17	−12.56	91.44
GDP growth	213	3.67	3.08	−8.98	15.23
Internet users	208	62.98	28.38	1.04	100.00
Mobile subscriptions	203	113.74	34.21	24.79	319.49
Broadband subscriptions	199	16.86	15.16	0.00	47.80
Log secure servers	214	6.43	3.25	0.23	12.53
Account ownership	200	59.78	28.21	3.66	99.98

Note. Unit of observation is a country-year survey. Account ownership is shown after carrying the latest Findex wave forward within country. India contributes one outcome observation (2022).

3.2. Models

Following the outcome-based tradition of Kuntchev et al. (2013), the *baseline* explains constraint with traditional financial-depth and macro variables only, fitted by ridge regression (Hoerl & Kennard, 1970) with a cross-validated penalty on standardised inputs. We then make two improvements: (i) we add the five digital-access indicators, and (ii) we replace the linear model with tree ensembles. The random forest (Breiman, 2001) uses 300 trees with a minimum leaf size of 3. The gradient boosting model (Friedman, 2001) uses 150 shallow trees (depth 2), a learning rate of 0.05 and a subsample fraction of 0.8. Models are implemented in scikit-learn (Pedregosa et al., 2011) with a fixed random seed of 42. The four specifications are therefore: baseline ridge, ridge with digital features, random forest with digital features, and gradient boosting with digital features.

3.3. Validation

Because the same economy can appear in several survey years, a random split would leak information between training and test sets. We therefore use 5-fold GroupKFold cross-validation by country, so every score is out-of-country. We report out-of-sample R^2 and mean absolute error (MAE, in percentage points).

3.4. Interpretation and the India counterfactual

Variable importance is measured by permutation importance (30 repeats) for the best-performing model. For India, the models are refit without India and used to predict its constraint. A counterfactual then lifts each of India's digital indicators to the sample 75th percentile (and leaves them unchanged where India is already above it). Because digital reach and income per capita are highly collinear, the counterfactual is an associational benchmark of headroom, not a causal effect.

4. Findings

4.1. India in context (RQ1)

In 2022, 34.5% of Indian firms are credit constrained (10.5% fully and 24.0% partially). This is above the sample median of 26.7%, and 36% of survey observations are more constrained than India. Fig. 1 compares India with emerging-market peers. Between 2014 and 2022 the share of Indian firms using banks to finance investment rose from 30.3% to 56.1%, internet use from 13.5% to 55.9%, and account ownership from 53.1% (2014) to 77.5% (2021), while private credit to GDP fell from 51.9% to 38.2%. Digital reach therefore rose sharply while bank-credit depth fell.

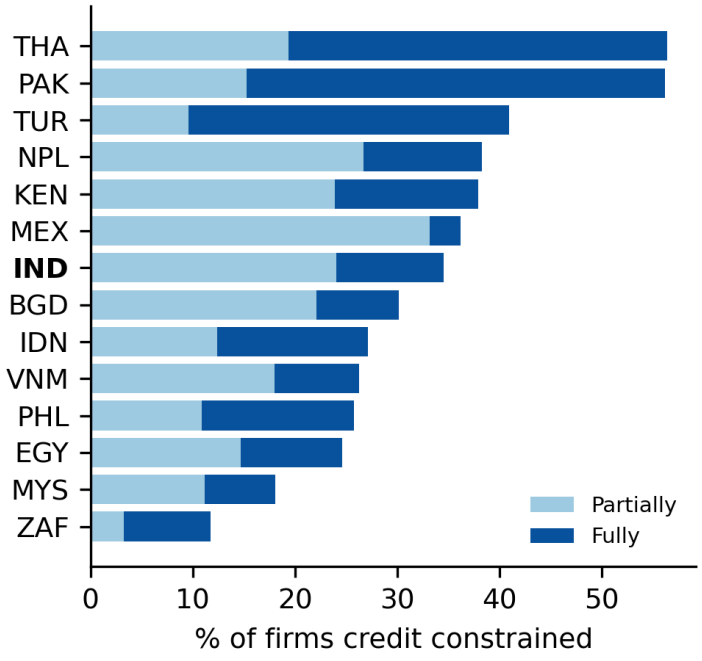


Fig. 1. Share of credit-constrained firms, India and selected peers (latest survey). India is shown in bold.

4.2. Digital access and constraint (RQ2)

Digital access tracks lower constraint. Internet use correlates with the share of constrained firms at $r = -0.59$ ($n = 208$), and account ownership at $r = -0.57$ ($n = 200$) (Fig. 2). India sits within the cloud of observations rather than at an extreme.

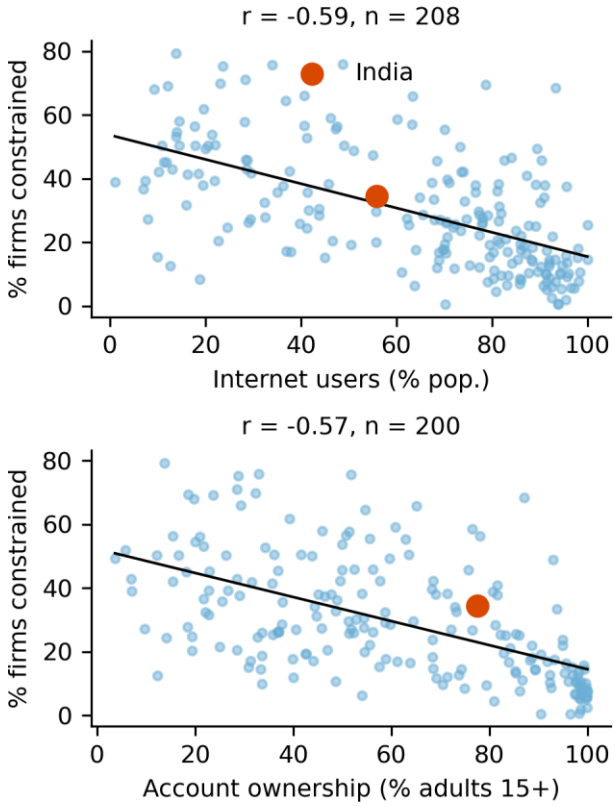


Fig. 2. Constraint versus digital access, with India highlighted. The line is a linear fit.

Table 5 and Fig. 3 show whether digital features improve prediction. Adding them to the linear model changes almost nothing (R^2 0.505 to 0.506), but the tree ensembles lift R^2 to 0.52 and cut MAE by up to 0.3 percentage points. The improvement is small. We do not claim digital access is a strong independent driver once income is accounted for, since digital reach and GDP per capita are highly collinear.

Table 5

Country-held-out cross-validation results (5 folds)

Model	R^2	MAE (pp)
Baseline: ridge (financial depth)	0.505	10.16
+ Digital: ridge	0.506	10.30
+ Digital: random forest	0.520	9.88
+ Digital: gradient boosting	0.522	9.99

Note. R^2 and MAE are computed on out-of-country predictions. pp = percentage points.

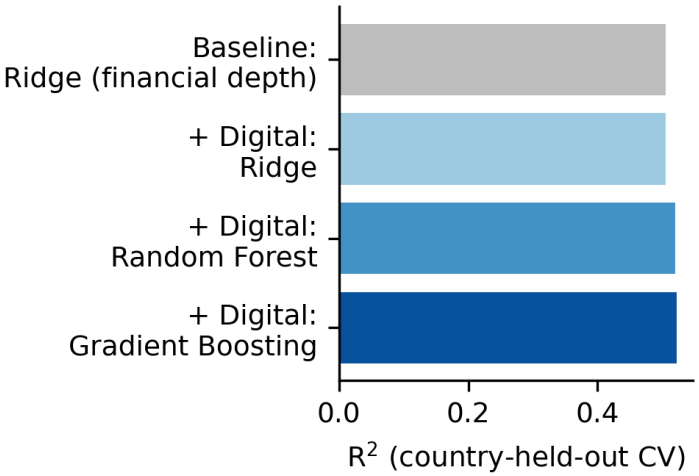


Fig. 3. Out-of-country R^2 by model.

4.3. Drivers of constraint (RQ2)

In the best model (gradient boosting, by R^2), log GDP per capita dominates, with a permutation importance of 0.34, followed by private credit to GDP (0.18). Digital variables matter jointly: secure servers (0.11), broadband (0.09), mobile subscriptions (0.06), account ownership (0.05) and internet use (0.05) (Fig. 4).

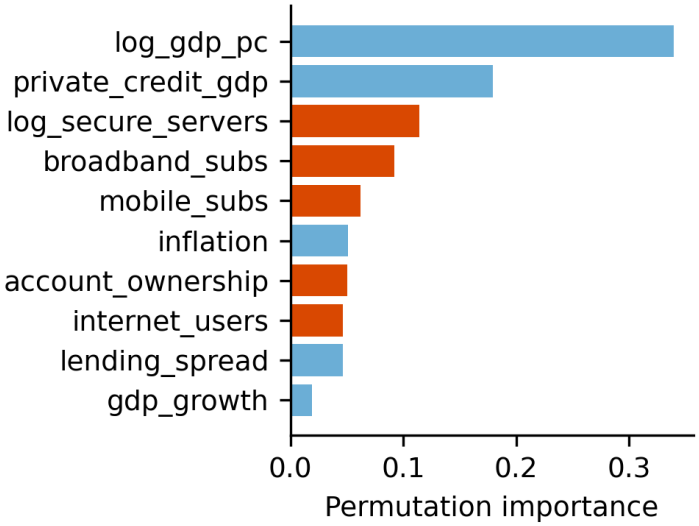


Fig. 4. Permutation importance in the gradient-boosting model. Digital features are shown in orange and traditional features in blue.

4.4. India: prediction and counterfactual

With India excluded from training, the baseline predicts 38.8% constrained against an actual 34.5%, whereas the digital gradient-boosting model predicts 31.8% (Table 6). The digital model therefore fits India better than the baseline does. Lifting India's digital indicators to the sample top quartile lowers the prediction to 27.3% (−4.5 pp). This is an associational benchmark and inherits the collinearity noted above. India also has only a single outcome observation (2022).

Table 6
India's constrained share: actual and out-of-sample predictions (2022)

Quantity	Share of firms constrained (%)
Actual	34.5
Baseline ridge, India excluded from training	38.8
Digital gradient boosting, India excluded from training	31.8
Counterfactual: digital indicators at sample top quartile	27.3

5. Discussion

5.1. Answers to the research questions and conclusions drawn

This subsection returns to the questions posed in Section 1 and states what the evidence does and does not allow us to conclude about each (Table 7).

RQ1: How constrained are Indian firms relative to peer economies? In 2022, 34.5% of Indian firms were credit constrained (10.5% fully, 24.0% partially), against a sample median of 26.7%, and 36% of survey observations were more constrained than India. *Conclusion:* India is more constrained than the typical economy in the sample, as Fig. 1 shows against emerging-market peers. The result is a descriptive benchmark based on a single India observation, so we cannot say whether the constraint is rising or falling. Over 2014–2022 India's digital reach and bank use by firms rose sharply while private credit to GDP fell, so the headline level hides a shift in how credit reaches firms.

RQ2: Do digital-access indicators explain credit constraint beyond traditional financial depth? Digital access is strongly associated with lower constraint in raw terms (internet use $r = -0.59$; account ownership $r = -0.57$). However, once income and financial depth are in the model, the additional explanatory power is small: out-of-country R^2 moves from 0.505 to 0.522 and MAE from 10.2 to 10.0 percentage points, and in the linear model the gain is negligible (0.505 to 0.506). Income per capita dominates (permutation importance 0.34), followed by private credit to GDP (0.18), while the five digital variables jointly carry a meaningful share (0.05–0.11 each). *Conclusion:* digital access goes with fewer constrained firms, but it is largely indistinguishable from general economic development in these data. We therefore answer "yes, modestly" and decline to call digital access a strong independent driver, because digital reach and GDP per capita are highly collinear.

RQ3: What does a simple machine-learning model imply for India's position and its digital headroom? With India held out of training, the traditional baseline predicts 38.8% constrained and the digital model 31.8%, against an actual 34.5%. Raising India's digital indicators to the sample top quartile lowers the prediction to 27.3% (−4.5 pp). *Conclusion:* the digital model describes India better than the baseline, which is consistent with India's public digital infrastructure substituting in part for collateral and credit depth, and the counterfactual points to several points of headroom. Both statements are associational. The counterfactual is a benchmark of what comparable economies look like, not an estimate of what a policy would achieve.

Central question: does digital access go with fewer credit-constrained firms, and where does India stand? Taken together, the answer is that digital access does go with fewer constrained firms across economies, India stands above the median level of constraint but better than its traditional fundamentals predict, and the evidence supports association, not causation.

Table 7
Summary of research questions, answers and strength of evidence

Question	Answer from the evidence	Strength of evidence
RQ1: India vs peers	34.5% of Indian firms constrained (2022) vs median 26.7%	Descriptive; one India observation
RQ2: Digital beyond financial depth	Strong raw correlation ($r \approx -0.6$) but small gain (R^2 0.505 to 0.522)	Associational; collinear with income
RQ3: Implication for India	Model predicts 31.8% vs baseline 38.8%; top-quartile digital reach gives 27.3%	Benchmark of headroom, not causal

5.2. Academic implications

First, the study shows that credit constraint can be benchmarked directly from outcome-based, openly available data, which answers the reliance of the literature on KZ and WW index proxies (Kaplan & Zingales, 2000; Whited & Wu, 2006). Second, it extends evidence that has concentrated on Chinese listed firms (Han & Zhang, 2023; Yao & Yang, 2022) to India and a global peer set. Third, it adds a cautionary result: digital indicators are strongly correlated with constraint ($r \approx -0.6$) but add only modestly to out-of-country prediction once income and financial depth are accounted for. That is what one would expect if digital reach largely reflects development, and it warns against reading raw correlations as evidence of a digital effect. Fourth, the work illustrates how a simple, transparent machine-learning comparison can complement econometric designs.

5.3. Managerial implications

For entrepreneurs and SME managers, digitising accounts and payments improves lender visibility, which is the channel through which digital finance is thought to work (AIM RNP, 2019). Firms that route sales through digital payments, keep machine-readable books and file tax returns create the data trail that alternative-data underwriting needs. For lenders, alternative-data underwriting should be designed so as not to exclude firms with thin digital footprints, who may be creditworthy but invisible. Banks should also consider data-sharing partnerships with FinTechs, which supply data and speed while banks supply capital (Larsson et al., 2024).

5.4. Policy implications

India's model-implied position is better than the traditional baseline suggests (31.8% predicted against 38.8%), which is consistent with digital public infrastructure doing some of the work that collateral and credit depth do elsewhere. At the same time, private credit to GDP fell from 51.9% to 38.2% over 2014–2022, which signals room for policy to strengthen bank–FinTech data sharing, for example through broader use of consent-based data portability under the Account Aggregator framework. Policy should also watch falling credit depth and ensure that digital rails reach firms that are currently outside them. The counterfactual (−4.5 pp) suggests several points of headroom, but should be read as a benchmark and not a forecast of policy impact.

5.5. Limitations

Four limitations qualify the results. (1) Country-level aggregates (n = 215) support only ecological, associational conclusions, not statements about individual firms or causal effects. (2) India has one outcome observation (2022), so no time series is possible. (3) Digital indicators are population-level and do not measure firm-level digital credit usage. (4) Global Findex values are carried forward between triennial waves, and the sample is unbalanced across years: 95 of the 215 observations come from 2023–2024. Digital reach and income are also highly collinear, which limits how far their separate roles can be identified.

5.6. Future scope

Future work should use the firm-level India Enterprise Survey microdata, which would allow constraint to be linked to firm-level digital behaviour, and state-level UPI and credit data, which would add the time and regional variation that the single India observation lacks. Quasi-experimental designs, such as staggered digital-lending regulation or the roll-out of the Account Aggregator framework, would move the analysis from association towards causal identification. Extensions could also test whether the digital premium differs for young, asset-light and first-time borrowers, the groups at the centre of the motivating argument.

6. Conclusion

Using open data, this study measured credit constraint directly and found that 34.5% of Indian firms were constrained in 2022, above the sample median of 26.7%. Digital-access indicators correlate strongly with lower constraint, but they add only modest out-of-country predictive power over traditional financial depth (R^2 from 0.505 to 0.522), with income per capita remaining the dominant predictor. A digital counterfactual suggests several points of headroom for India (27.3% against a model-implied 31.8%). The evidence is associational: firm-level and causal work remains the priority. For policy-makers, managers and lenders, the practical message is to build on India's public digital infrastructure through data sharing and digitised firm records, while keeping thin-footprint firms in the credit net.

Data availability

All data are openly available from the World Bank API (https://api.worldbank.org). The raw panel, the analysis notebook (*analysis.ipynb*) and all figures are in the project folder and reproduce every number reported here.

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